TSA 2014

Problem Solving

Questions taken from the TSA Oxford 2014 Section 1 question paper

Questions	Total marks	Time limit
25	25	45 minutes

Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Questions are numbered 1 to 25 for this set. The number in brackets shows the question's number in the original TSA 2014 paper, e.g. **1 (Q3)**.

How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to the 25 questions in this set: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes).

Score: _____ / 25

Questions © UCLES 2014. Reproduced from the TSA Oxford 2014 Section 1 question paper for practice purposes.

1 (Q2) Two neighbours work for the same company and share the journey to work, driving alternately in strict rotation. They work Monday to Friday each week and every other Saturday. (They always work the same Saturdays as each other.)

What is the **maximum** number of days either of them has to drive in a calendar month?

A 11

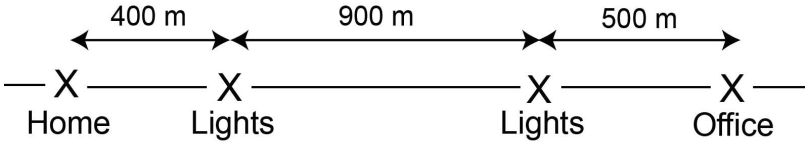
B 12

C 13

D 14

E 15

2 (Q6) Central Avenue is a long straight road, several kilometres long. Both my home and my office are on Central Avenue, and the diagram below shows their positions, together with two sets of traffic lights that lie in between.



Both sets of lights turn to green for Central Avenue traffic simultaneously, showing green for exactly two minutes, turning green again exactly two minutes later.

I cycle to work every morning at a steady speed of 5 metres per second (except, of course, whilst waiting at traffic lights).

What is the **longest** journey time I can expect the journey to work to take?

A 6 minutes

B 7 minutes

C 8 minutes

D 9 minutes

E 10 minutes

© UCLES 2014

3 (Q7) The United States Postal Service has an online guide to help people using their postal service. This is an extract:

First Class Mail

First class mail must be used for handwritten or personal correspondence.

First Class Mail Rates – single piece rates

First 30 grams Letter size mail	$0.44
Each additional 30 grams or part thereof	$0.17 up to 360 grams
First 30 grams Parcel	$1.22
Each additional 30 grams or part thereof	$0.17
Standard Postcard	$0.28

Letters: 30 cm long, 16 cm wide, 1 cm thick, maximum weight 360 grams. (Any item larger than this must be sent as a parcel).

Parcels: Dimensions cannot exceed 1250 cm.

I want to send two items through the post using first class mail. One is 50 cm long, 25 cm wide, 10 cm thick and weighs 300 grams. The other is 20 cm long, 12 cm wide, 0.20 cm thick and weighs 110 grams.

How much will it cost me to send the items?

A $2.92

B $3.53

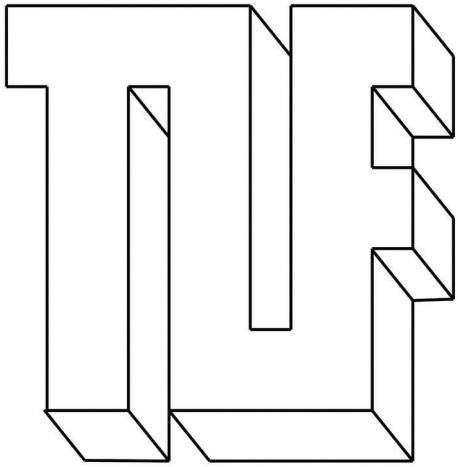
C $3.70

D $4.04

E $4.48

© UCLES 2014

4 (Q8) Below is Thomas Leslie Fuller's brass paperweight, which shows his initials.

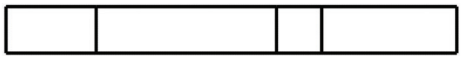


Which one of the following is **not** a side view of Thomas' paperweight when it is placed flat on a table (either side up)?

A



B



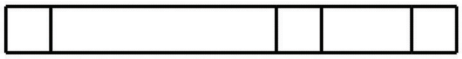
C



D



E



© UCLES 2014

5 (Q12) David is planning a holiday for himself and his wife. He has a budget of £2,000 but wishes to retain about £600 for spending money. Below is a table showing different holiday possibilities.

	Return flight	**Ferry and train**	**Car hire and fuel**	**Hotel accommodation**
Moscow	£300 per person	£250 per person	N/A	£120 per night for 2 only
Cyprus	£195 per person	N/A	N/A	£42.50 per person per night
Barcelona	£215 per person	£210 per person	N/A	£370 per week per person
Paris	£80 per person	£78 per person	N/A	£180 per night per room for 2
Scotland	N/A	N/A	£40 per day	£80 per night for 2 only
Rome	£110 per person	£220 per person	N/A	£120 per night for 2 only

Which one of the following holiday possibilities leaves him with as near to £600 as possible?

A Ten nights in Moscow, travelling by ferry and train.

B Twelve nights in Cyprus.

C Twelve days and nights in Scotland.

D Twelve nights in Rome travelling by plane.

E Seven nights in Paris travelling by plane.

© UCLES 2014

6 (Q13) The table below shows the energy values for the main foods used by volunteer wardens at a wild bird reserve to prepare a variety of feed mixes.

Food	Energy per 100 g (calories)
mealworms	150
peanuts	560
apples	350
oats	370
niger seed	480
cheese	400
sunflower seeds	500
raisins	300
sunflower hearts	600
suet	800

A trainee volunteer has been asked to prepare a sample of a special mix to provide exactly 5,000 calories. He has weighed out the first four ingredients but cannot read the required mass of the final ingredient because of an ink blot. This is what has been weighed out so far:

mealworms 150 g, apple 150 g, raisins 250 g, suet 125 g

What mass of sunflower seeds must be used to successfully complete the task?

A 500 g

B 520 g

C 540 g

D 560 g

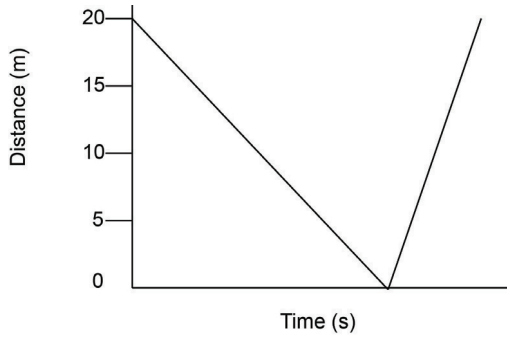
E 680 g

© UCLES 2014

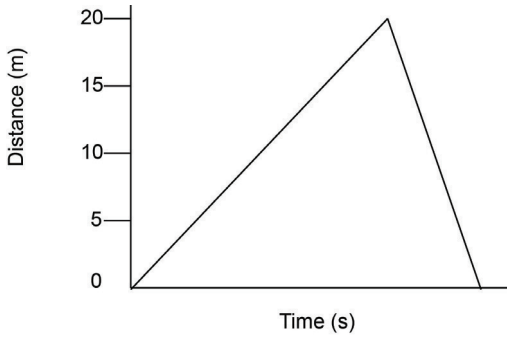
7 (Q14) Sue takes her dog Freya to the park. She throws a stick for Freya to fetch. Freya runs to collect the stick and bring it back to Sue. While Freya is collecting the stick, Sue walks slowly towards Freya.

Which one of the following graphs could correctly show the distance between Sue and Freya?

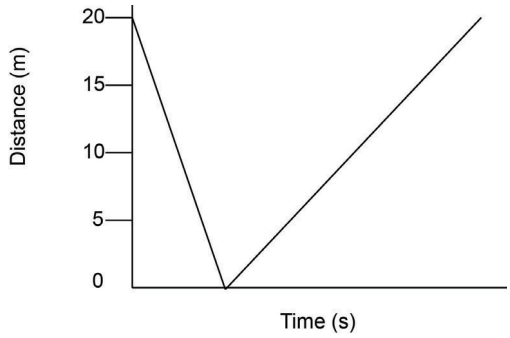
A



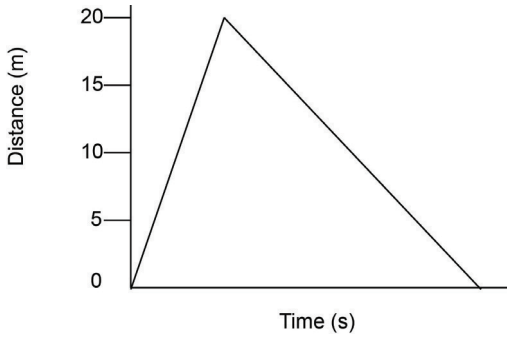
B



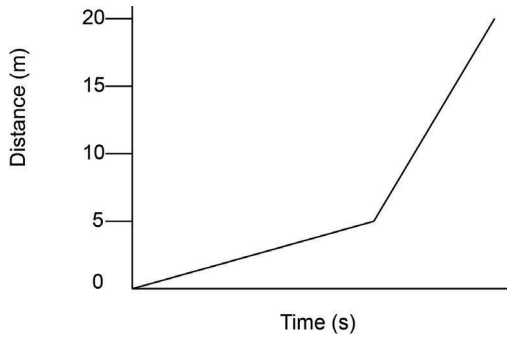
C



D



E



© UCLES 2014

8 (Q18) A main road crosses a side road and the junction is controlled by a set of traffic lights. The normal sequence is for the lights to be green for 25 seconds on the main road followed by 15 seconds of green for the side road. There is also a pedestrian phase in the lights when all the road lights are red. If a pedestrian presses a button to cross any of the roads then, after the side road lights have been green, the pedestrian lights turn green for 10 seconds. Following a pedestrian phase, the main road green period is increased to 35 seconds to clear the build-up. Whenever any of the lights change there is a two-second period when all lights are red. The time on amber is included in the green times given above.

What is the longest time traffic on the side road will have to wait for their lights to go green?

A 19 seconds

B 29 seconds

C 45 seconds

D 51 seconds

E 66 seconds

9 (Q19) Elsie has six grandchildren. Their birthdays are as follows:

Ryan	19th February
Rebecca	3rd May
Richard	30th May
Ralph	7th September
Ruth	17th November
Robert	16th December

Remarkably, in non-leap years the boys' birthdays are on the 50th, 150th, 250th and 350th days of the year, whilst the girls' birthdays are on the 123rd and 321st days, and all six are on different days of the week.

Which other grandchild's birthday is on the same day of the week as Ryan's in a leap year?

A Ralph

B Rebecca

C Richard

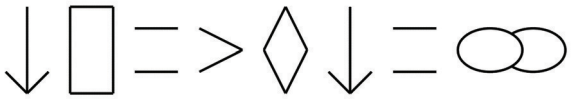
D Robert

E Ruth

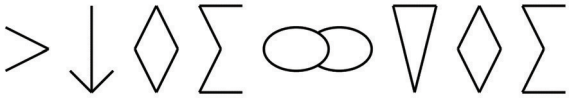
© UCLES 2014

10 (Q20) Whenever my friend Alistair writes to me he does so entirely in code. He always uses symbols for letters but changes the code each time, so any particular symbol does not necessarily stand for the same letter of the alphabet every time.

I always start to crack the code by looking at how he has written his name at the end of his letter. For instance, last time he was:



In my last letter to him I asked him when he intends to visit me next. Today I have received a reply consisting of just one word as follows:



When can I assume that he intends to appear?

A TOMORROW

B THURSDAY

C SATURDAY

D NOVEMBER

E SOMETIME

© UCLES 2014

11 (Q24) A hotel offers its guests a dinner allowance of £28.50 per night. A guest wishes to spend up to her allowance on a three course meal without overspending.

Here is the menu:

STARTER	PRICE (£)
Soup of the day	5.00
Prawn cocktail	6.50
Smoked salmon	6.10
Duck confit salad	6.95
MAIN COURSE	
Sirloin steak 8 ozs.	14.90
Gammon steak	12.50
Rainbow trout	16.10
Duck a l'orange	15.15
Vegetarian risotto	12.75
DESSERT	
Cheese and biscuits	6.10
Cheesecake	5.60
Apple pie and cream	5.30
Lemon sorbet	5.20
Profiteroles	6.20

What is the largest sum she can spend without exceeding the limit?

A £28.05

B £28.30

C £28.35

D £28.40

E £28.45

© UCLES 2014

12 (Q25) Chris is an actor and recently feels he has become overweight. He aims to lose at least 7.5 lbs over the next 10 weeks. He decides he is able to walk the following number of miles over each of the next ten weeks:

Mondays – 5 miles

Wednesdays – 5 miles

Saturdays – 8 miles

Sundays – 8 miles

The table below shows calories burned per mile by walking at various speeds.

		Starting Weight								
		100lb	120lb	140lb	160lb	180lb	200lb	220lb	250lb	275lb
Walking speed	2.5mph	55	65	76	87	98	109	120	136	150
	3.0mph	53	64	74	85	95	106	117	133	146
	3.5mph	52	62	73	83	94	104	114	130	143
	4.0mph	57	68	80	91	102	114	125	142	156
	4.5mph	64	76	89	102	115	127	140	159	175
	5.0mph	73	87	102	116	131	145	160	182	200

NB: A lb in fat equals 3500 calories

What is the slowest speed he needs to walk at to lose at least 7.5 lbs if he starts at 180 lbs?

A 2.5 mph

B 3.0 mph

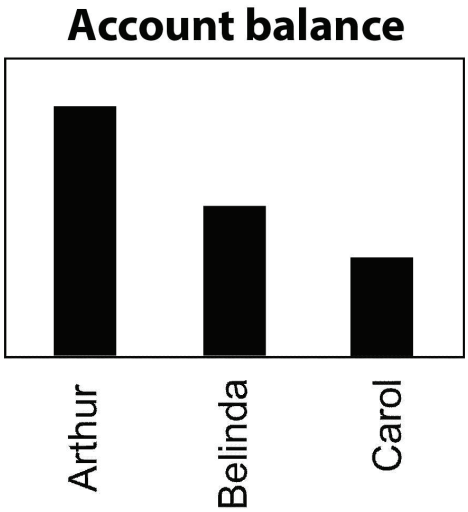
C 4.0 mph

D 4.5 mph

E 5.0 mph

© UCLES 2014

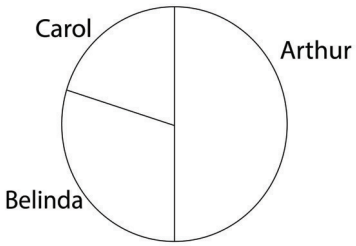
13 (Q26) The bar chart shows the amount of money in the savings accounts of three friends at the start of the year.



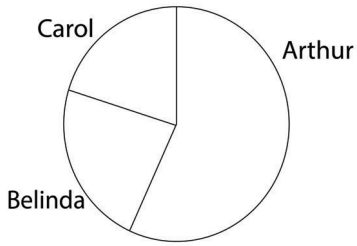
Since then, they have all spent the same amount of money from their accounts and the amount in Belinda's account has halved.

Which one of the following pie charts could represent the amounts of money in the accounts now?

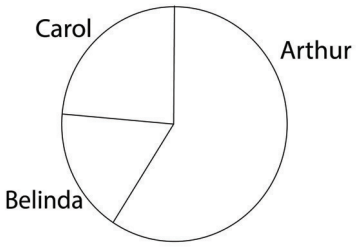
A



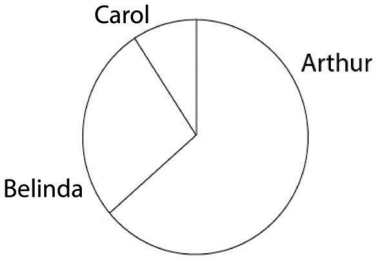
B



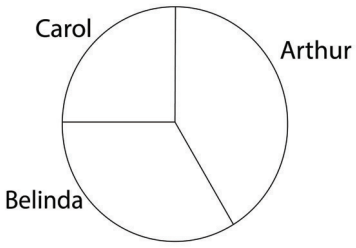
C



D



E



© UCLES 2014

14 (Q30) David has a new cycle computer but has not learned all the functions yet, so can only read the cumulative mileage and average speed of all of the journeys since he fitted it. On his first journey using it he covered 15 km at an average speed of 30 km/h. He can work out from this that the journey took him 15/30 hours or exactly 30 minutes. After the second journey, his cumulative mileage was 24 km and overall average speed 32 km/h.

What was his average speed on the second journey?

A 18 km/h

B 31 km/h

C 34 km/h

D 36 km/h

E 45 km/h

15 (Q31) I have bought a combination lock for my bicycle. The combination is a three-digit number in the range 000-999 and it is currently set to the manufacturer's default combination of 999. Before I use the lock I need to set the combination. I want to use a combination where all three digits are different.

How many combinations are there in the range 000-999 where all three digits are different?

A 504

B 720

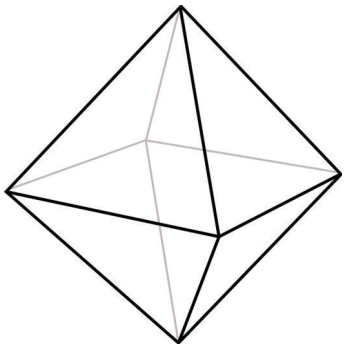
C 810

D 900

E 990

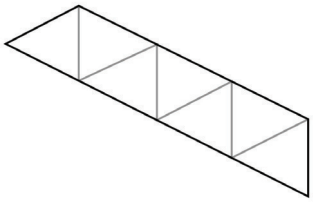
© UCLES 2014

16 (Q32) A regular octahedron is a solid composed of eight equilateral triangles, as shown in the image below:

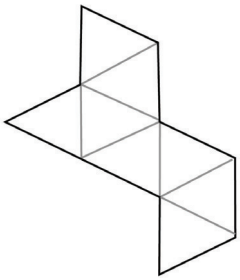


Which one of the following paper shapes will fold into an octahedron?

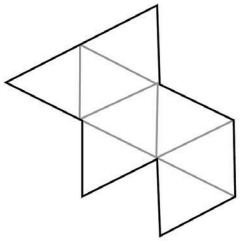
A



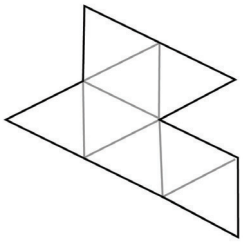
B



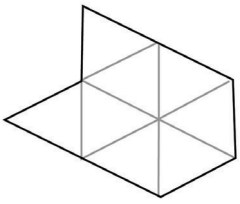
C



D



E



© UCLES 2014

17 (Q36) Lucy allowed an hour to travel to a job interview, intending to arrive exactly on time. The train which she took arrived ten minutes late by her watch. The taxi which she then took got caught in a traffic jam for an unexpected 15 minutes and walking to the appropriate room took five minutes longer than anticipated. The clock in the interview room was running 30 minutes slower than Lucy's watch which was running ten minutes fast.

According to the clock in the interview room, when did Lucy arrive?

A She was ten minutes early.

B She was ten minutes late.

C She was thirty minutes late.

D She was sixty minutes late.

E She was on time.

18 (Q37) My current membership of the squash club is about to expire and so I wish to renew it. The options available to me are given in the table below and have not changed since I joined as a new member six months ago (a change in membership type counts as a renewal, not a new member).

Membership type	Length	Extras	New member (£)	Renewal price (£)
Bronze	1 year	none	10	8
Silver	6 months	free locker	25	18
	1 year	free locker	45	28
Gold	6 months	free locker free competition entry	40	32
	1 year	free locker free competition entry	75	52

I am currently a Gold member, but I do not want to enter competitions any more as I play only once a month. I do need a locker when I visit and the normal cost to hire a locker is £2 per visit. I want to choose a membership that will give me the smallest possible total cost for a full year.

What will be the difference between the amount that I will pay now and the amount I paid six months ago?

A I will pay £32 less.

B I will pay £30 less.

C I will pay £12 less.

D I will pay £8 less.

E I will pay £5 more.

© UCLES 2014

19 (Q38) The table shows the number of children in the town of Lancester aged 11 and 16 who play various sports after school.

Sport	11-year-olds	16-year-olds
Football	120	181
Cricket	120	133
Hockey	55	66
Swimming	104	150
Tennis	123	149
Squash	51	97

In which sport was the proportion of 11-year-old to 16-year-old children nearest to that for swimming?

A Football

B Cricket

C Hockey

D Tennis

E Squash

© UCLES 2014

20 (Q42) The table below shows the mean maximum daily temperature for Oxford for each year from 1905 to 1920 along with a moving 5-year average (the value shown beside 1920 is the average from 1916 to 1920 inclusive). Three values are missing from the table.

Year	Mean maximum daily temperature	5-year moving average
1905	13.4	13.40
1906	14.4	13.56
1907	13.4	13.60
1908	13.7	13.66
1909	x	y
1910	13.3	z
1911	14.9	13.66
1912	13.7	13.72
1913	14.2	13.82
1914	14.7	14.16
1915	13.6	14.22
1916	13.3	13.90
1917	12.8	13.72
1918	13.8	13.64
1919	13.0	13.30
1920	13.8	13.34

What value should be in the position marked 'y'?

A 13.30

B 13.64

C 13.58

D 13.66

E 13.72

© UCLES 2014

21 (Q43) Fred is carrying out a three-year biological survey which depends on anglers sending in scale samples of fish that they catch. The survey covers five rivers and Fred has set a target for the numbers of sets of fish scales returned from each river each year. Unfortunately, by the end of the second year the number of scale sets returned has fallen well short of target.

River	Annual target of scale sets	Scale sets obtained		Number of anglers sending scale sets		Percentage of target achieved	
		Year 1	*Year 2*	*Year 1*	*Year 2*	*Year 1*	*Year 2*
Dark	250	77	125	5	30	31	50
Fare	100	85	35	20	8	85	35
Gwynt	500	71	132	19	2	14	26
Tine	250	105	66	43	19	42	26
Yarrow	250	80	150	18	25	32	60

The anglers who participated in the survey sent scales from all the fish they caught.

For which river was the average catch per angler sending in scales the highest in any one year?

A Dark

B Fare

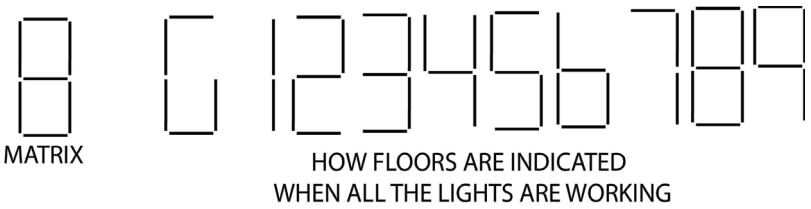
C Gwynt

D Tine

E Yarrow

© UCLES 2014

22 (Q44) Allan wishes to visit a client on floor 8 of a building with 10 floors (including the ground floor). He enters the lift on the ground floor. His client has explained by email that the floors are indicated by an illuminated matrix but there is a notice stating that two of the lights are not working.



MATRIX

HOW FLOORS ARE INDICATED
WHEN ALL THE LIGHTS ARE WORKING

Allan enters the lift and pushes the button. Just then he starts to sneeze, distracting his attention from the display. The lift leaves the ground floor, travels, then stops. He glances at the display and concludes he could be on floor 8 so he gets out. In fact he is on the wrong floor.

On how many of the floors could Allan have got out by mistake?

A 5

B 6

C 7

D 8

E 9

23 (Q48) Amy is researching her family history. She has two parents and four grandparents. However, she discovers that, among her great grandparents, two sisters from one family married two brothers from another.

How many great-great grandparents has she?

A 8

B 10

C 12

D 14

E 16

© UCLES 2014

24 (Q49) The West Water Canoe School offers tuition in basic canoeing skills. There are 18 canoes for students to use. The school offers a minimum instructor to student ratio of 1:6 and a minimum of two instructors for any group.

Instructors are hired by the centre on a casual basis and are paid only for the time they actually work. Instructors are paid £6 per hour and are expected to use their own canoes.

The cost for students depends on the numbers in the group and the length of the session. Costs are set out in the table below.

Number in group	Cost per student for first hour	Cost per student for subsequent hours
1 to 2	£12	£10
3 to 6	£10	£8
7 to 10	£8	£6
more than 10	£7	£5

What is the maximum profit the centre can make from a group after paying the instructors for a two-hour session?

A £108

B £180

C £198

D £216

E £360

© UCLES 2014

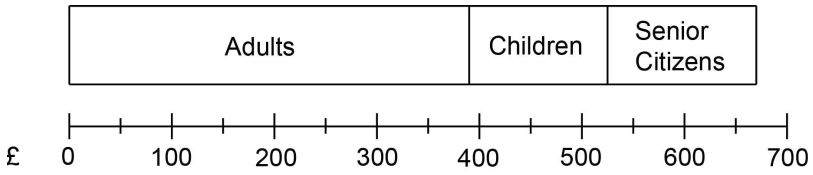
25 (Q50) The cost of entry to the Victoria and David Museum is as follows:

Adults £7

Children £4

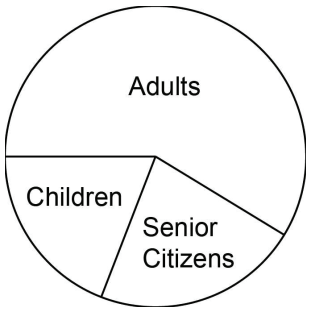
Senior Citizens £5

This chart shows the breakdown of yesterday's total takings of £679 by age group.

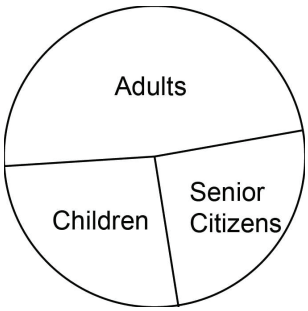


Which one of the following pie charts shows the number of visitors in each age group that paid to enter the museum yesterday?

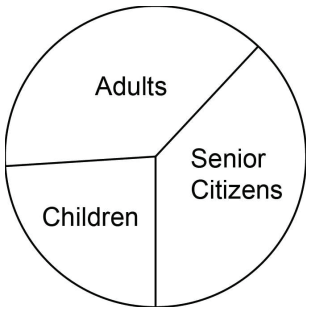
A



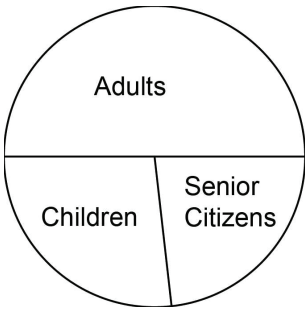
B



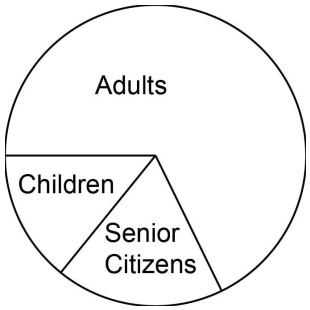
C



D



E



END OF TEST

© UCLES 2014